Activity 2.2: Triangles, Polygons & Parallel Lines

A

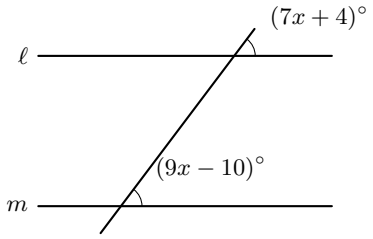
Name: _____ Partner's name: _____ Date: _____

Directions: Find the value or measure asked for. Show your work. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

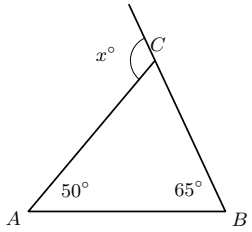
For what value of x are ℓ and m parallel?



Answer: _____ Match? ☐ Yes ☐ No

2

Find x.



Answer: _____ Match? ☐ Yes ☐ No

3

The angles of a triangle measure x°, $2x^\circ$, and $3x^\circ$. Find the measure of the largest angle.

Answer: _____ Match? ☐ Yes ☐ No

4

Find the measure of each base angle of an isosceles triangle whose vertex angle measures 40°.

Answer: _____ Match? ☐ Yes ☐ No

5

A regular polygon has exterior angles of 45°. How many sides does it have?

Answer: _____ Match? ☐ Yes ☐ No

6

Find the measure of each interior angle of a regular hexagon.

Answer: _____ Match? ☐ Yes ☐ No

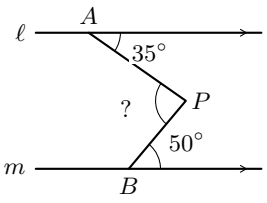
7

The angles of a triangle are in the ratio $2 : 3 : 4$. Find the measure of the smallest angle.

Answer: _____ Match? ☐ Yes ☐ No

8

$\ell \parallel m$. Find $m\angle APB$.



Answer: _____ Match? ☐ Yes ☐ No

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B

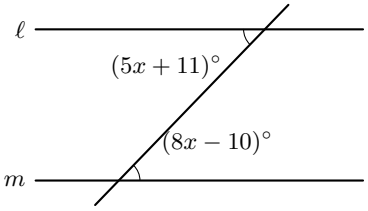
Name: _____ Partner's name: _____ Date: _____

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Work alone, then compare after each problem. If answers differ, find the mistake together.

1

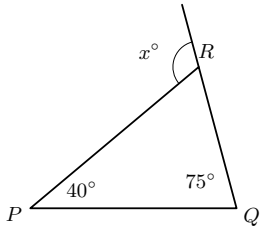
For what value of x are ℓ and m parallel?



Answer: _____ Match? ☐ Yes ☐ No

2

Find x.



Answer: _____ Match? ☐ Yes ☐ No

3

The angles of a triangle measure $x°$, $(x + 30)°$, and $(x + 60)°$. Find the measure of the largest angle.

Answer: _____ Match? ☐ Yes ☐ No

4

In an isosceles triangle, the vertex angle measures $30°$ less than each base angle. Find the measure of each base angle.

Answer: _____ Match? ☐ Yes ☐ No

5

The interior angles of a convex polygon add up to $1080°$. How many sides does it have?

Answer: _____ Match? ☐ Yes ☐ No

6

A regular polygon has exterior angles of $60°$. Find the measure of each interior angle.

Answer: _____ Match? ☐ Yes ☐ No

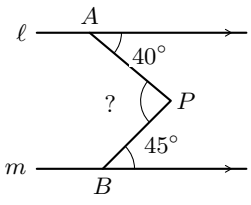
7

The angles of a triangle are in the ratio $4 : 5 : 9$. Find the measure of the smallest angle.

Answer: _____ Match? ☐ Yes ☐ No

8

$\ell \parallel m$. Find $m\angle APB$.



Answer: _____ Match? ☐ Yes ☐ No